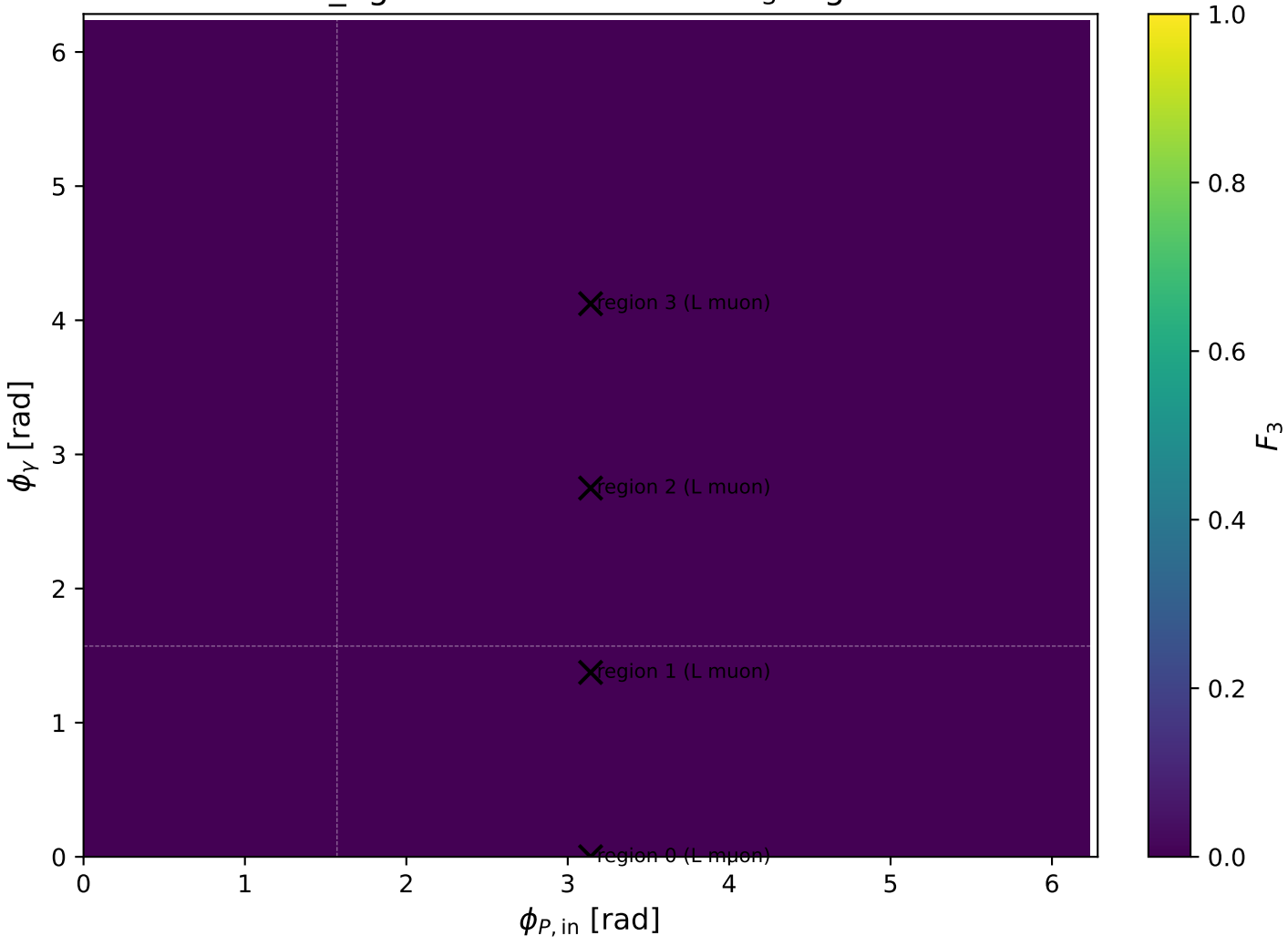
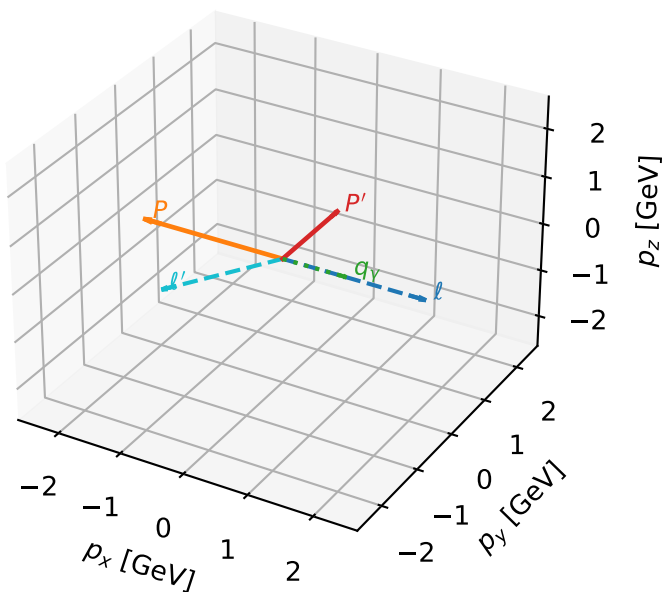


mid_Egamma L muon: max F_3 regions



L muon characteristic max F_3 configuration

3D momenta

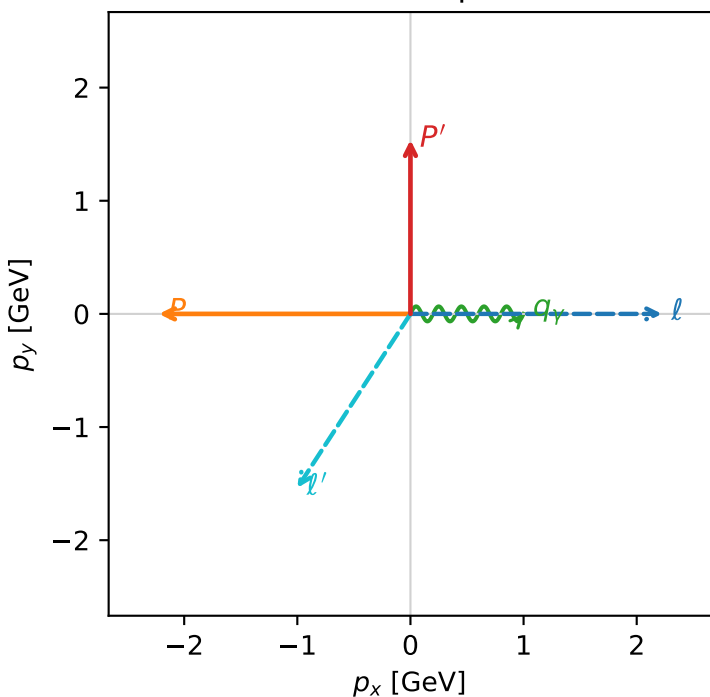


f3_mid_egamma_l_lepton_region_0 F_3=0
region: mid_Egamma_L_lepton_region_0
outgoing-state purity: 0.846761
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

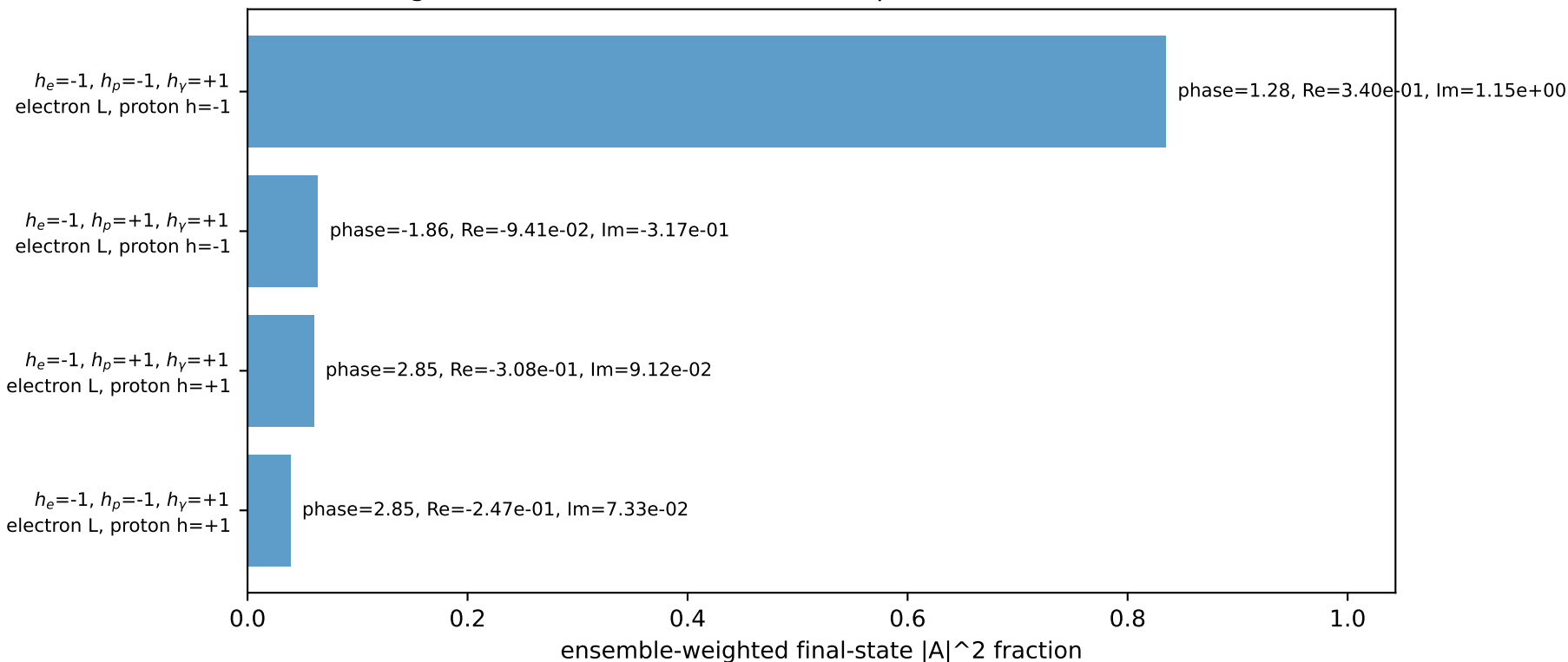
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.5377$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_Y=1$, $\phi_Y=0$
 $Q^2=12.6022$, $x_B=0.777691$, $t=-6.93259$, $W^2=4.48227$, $y=0.785684$
 $\theta(l', \gamma)=123.037$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.2231$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2231$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 1.8373$ $p_x= -1.0000$ $p_y= -1.5377$ $p_z= 0.0000$
 P' $E= 1.8012$ $p_x= 0.0000$ $p_y= 1.5377$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 1.0000$ $p_y= 0.0000$ $p_z= 0.0000$

Transverse plane

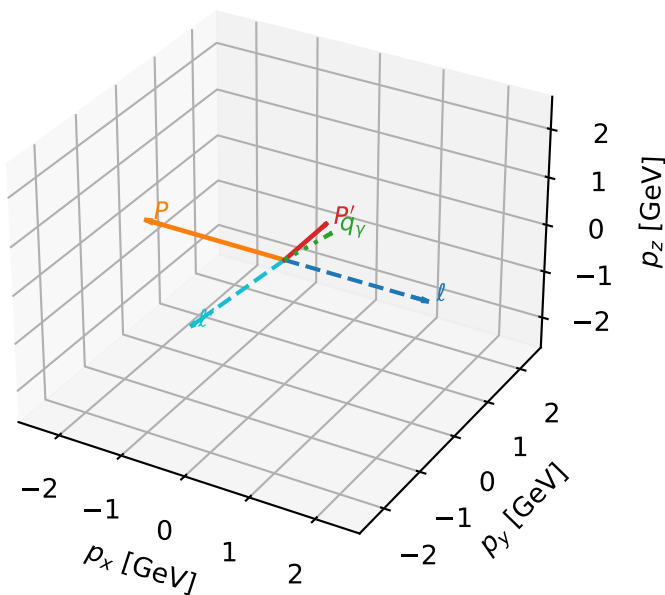


Leading initial-ensemble/final-state components (N=4, retained=99.8%)



L muon characteristic max F_3 configuration

3D momenta

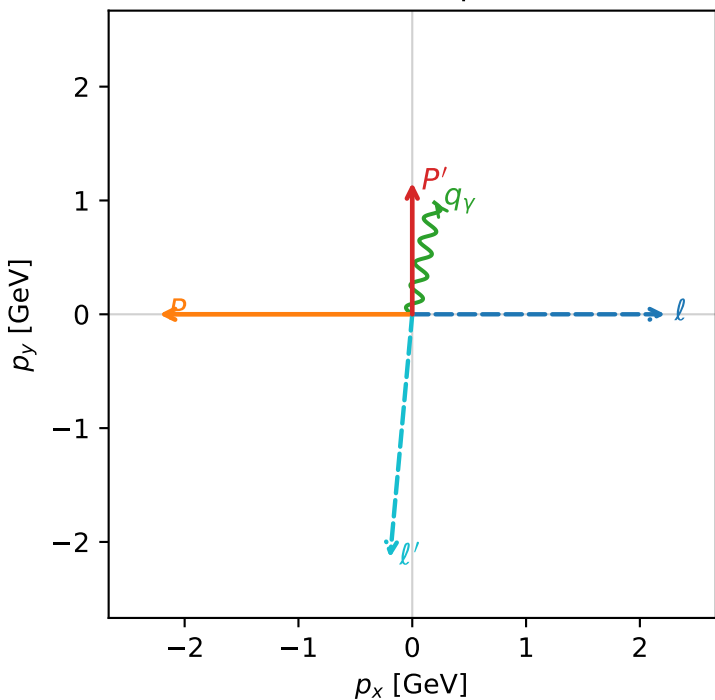


f3_mid_egamma_l_lepton_region_1 F_3=0
 region: mid_Egamma_L_lepton_region_1
 outgoing-state purity: 0.754643
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

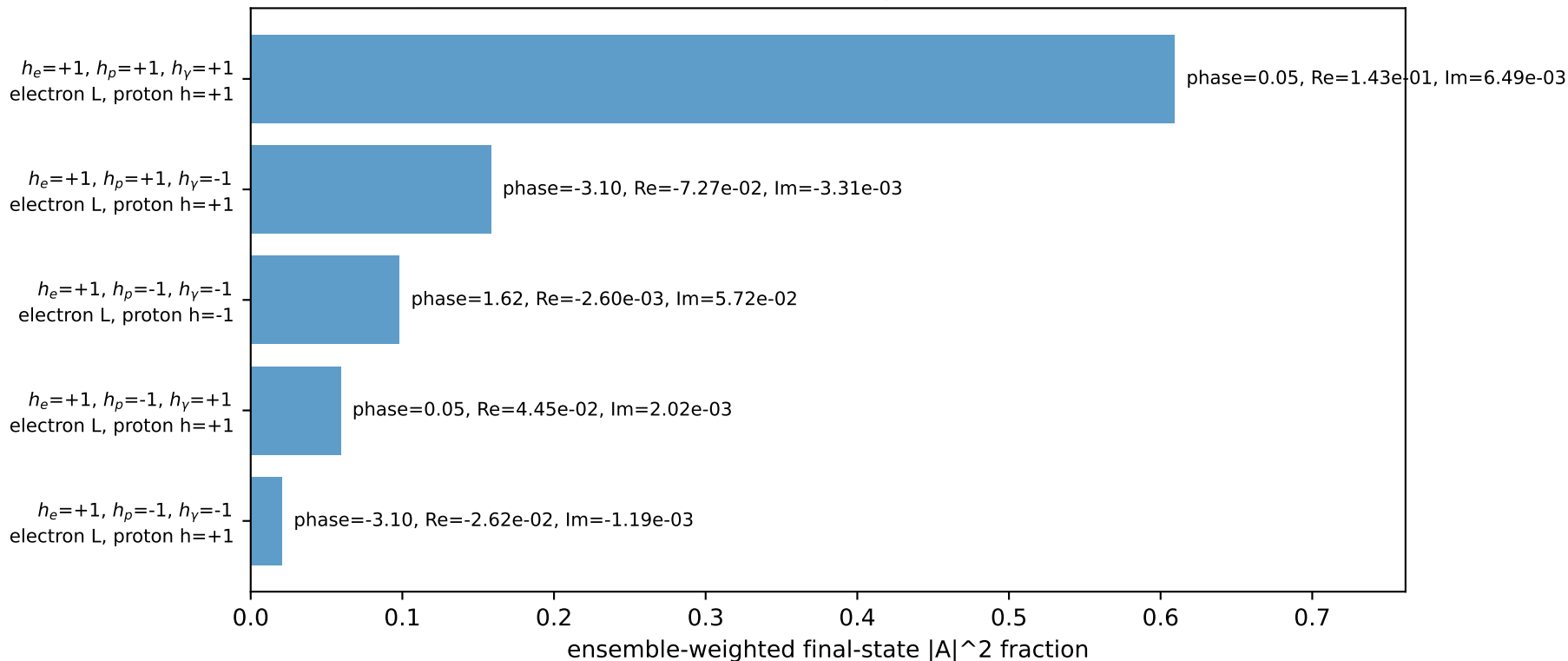
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.15688$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_Y=1$, $\phi_Y=1.37445$
 $Q^2=10.4115$, $x_B=0.936207$, $t=-5.4277$, $W^2=1.58928$, $y=0.5392$
 $\theta(l', \gamma)=173.965$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.2231$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2231$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 2.1491$ $p_x= -0.1951$ $p_y= -2.1377$ $p_z= 0.0000$
 P' $E= 1.4894$ $p_x= 0.0000$ $p_y= 1.1569$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.1951$ $p_y= 0.9808$ $p_z= 0.0000$

Transverse plane

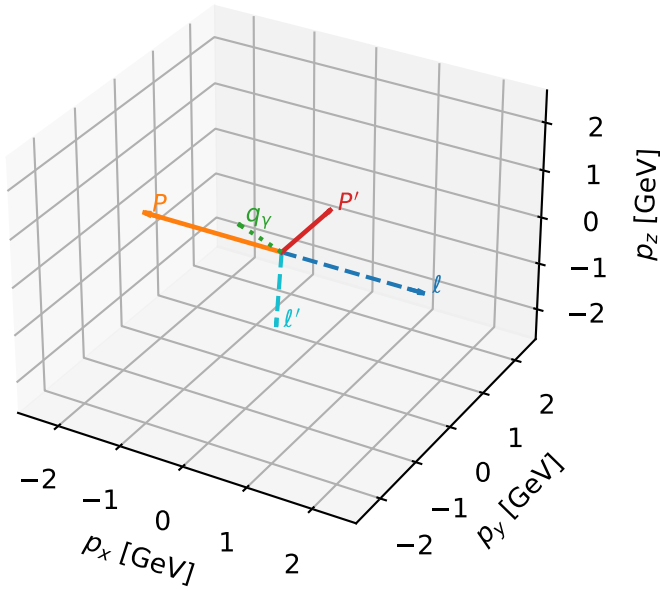


Leading initial-ensemble/final-state components (N=5, retained=94.5%)



L muon characteristic max F_3 configuration

3D momenta

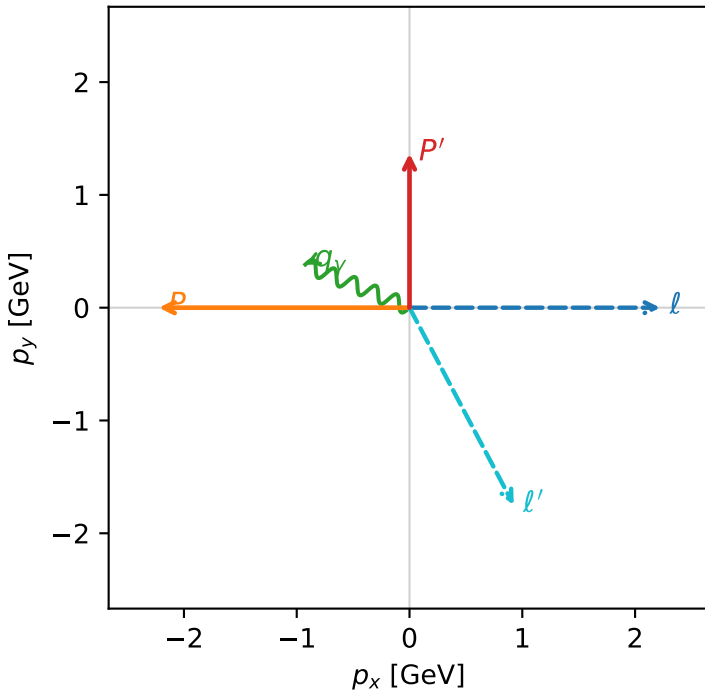


f3_mid_egamma_l_lepton_region_2 F_3=0
 region: mid Egamma_L_lepton_region_2
 outgoing-state purity: 0.606305
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_\mu, h_p)$

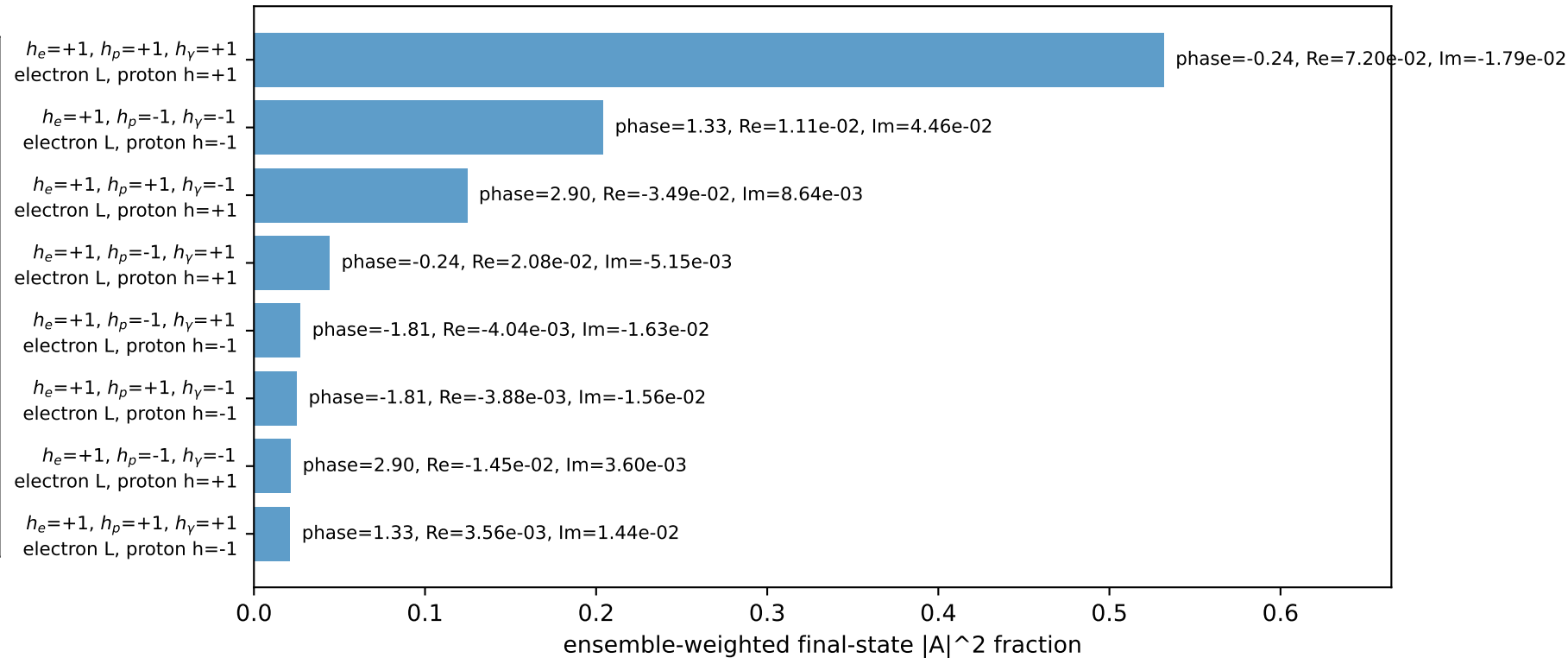
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.36654$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_\gamma=1$, $\phi_\gamma=2.74889$
 $Q^2=4.68795$, $x_B=0.673844$, $t=-6.239$, $W^2=3.14892$, $y=0.337313$
 $\theta(l', \gamma)=140.342$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.2231$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2231$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 1.9810$ $p_x= 0.9239$ $p_y= -1.7492$ $p_z= 0.0000$
 P' $E= 1.6575$ $p_x= 0.0000$ $p_y= 1.3665$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.9239$ $p_y= 0.3827$ $p_z= 0.0000$

Transverse plane

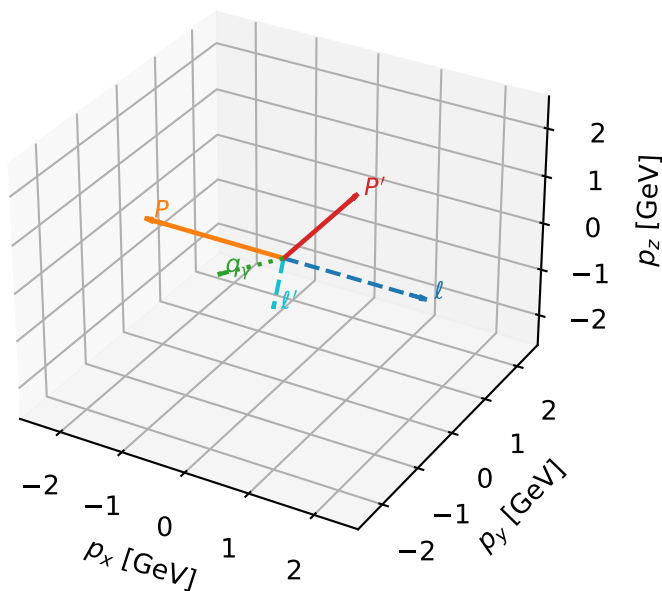


Leading initial-ensemble/final-state components (N=8, retained=100.0%)



L muon characteristic max F_3 configuration

3D momenta

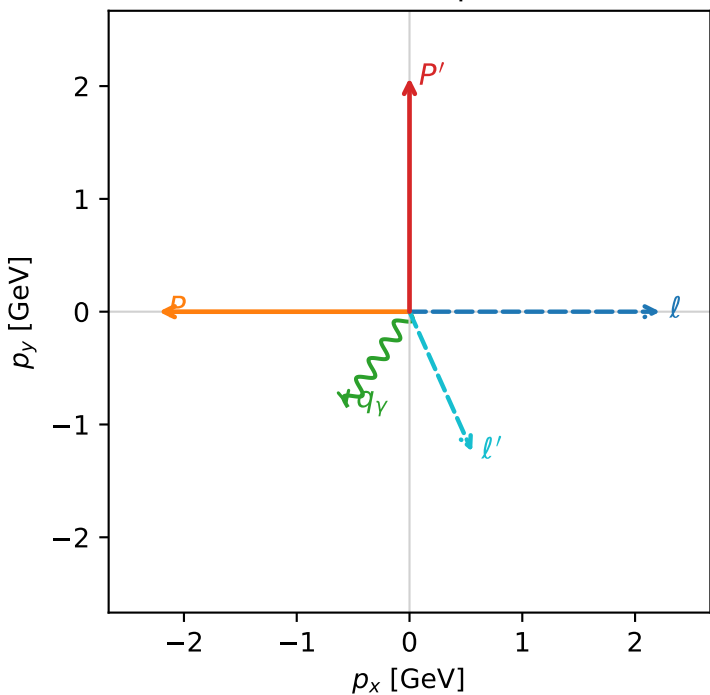


f3_mid_egamma_l_lepton_region_3 F_3=0
region: mid Egamma_L_lepton_region_3
outgoing-state purity: 0.663054
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\mu, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.0724$
 $\theta_{in}=1.57079$, $\phi_l=0$, $\phi_p=3.14159$
 $E_\gamma=1$, $\phi_\gamma=4.12334$
 $Q^2=3.57774$, $x_B=0.309129$, $t=-9.21801$, $W^2=8.87572$, $y=0.561149$
 $\theta(l', \gamma)=57.8682$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 2.2231$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2231$ $p_y= 0.0000$ $p_z= 0.0000$
 ℓ' $E= 1.3637$ $p_x= 0.5556$ $p_y= -1.2409$ $p_z= 0.0000$
 P' $E= 2.2748$ $p_x= 0.0000$ $p_y= 2.0724$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.5556$ $p_y= -0.8315$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=96.6%)

